



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	SE	Semester:	VIII
Course Code:		Course Name:	RL

Name of Student:	BART ANILK VZNOJ
Roll No. :	61
Assignment No.:	2
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	04
Demonstrated Knowledge	3	02
Legibility	2	02
Total	10	08

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : Ms. Sneha Yadav

Signature :

Date :

Q. 1) If youtube only shows you videos from channels you already subscribe to, find why that a "greedy" approach that hurts creators. Also identified how "exploration" helps the algorithm find a new viral video that you might like more than your current favorites.

→ Part 1: why showing only subscribed channels is a "greedy" approach - In reinforcement learning, a greedy policy always selects the action with the highest current

$$a = \operatorname{argmax}_a Q(s, a)$$

this means,

- always choose what seems best right now
- never try anything uncertain

- Applied to youtube -

- If youtube only shows videos from channels you already subscribe to:

- The algorithm assumes - "These channels gave high watch time"

- why this is greedy -

- maximizes immediately engagement
- ignores unknown creators
- never test new options

- why this hurts new creators -

- have no interaction history
- have low estimated reward
- never get shown
- never get clicks
- never get data to improve their ranking

- This creates a rich-get-richer effect.

- The system becomes trapped in a local maximum, it thinks current favorites are best, but it never checks if something better exists.

- Part 2 - How "exploration" helps find new viral videos
- In reinforcement learning, we use ϵ -greedy strategy with probability ϵ choose meaning.
- with probability $1 - \epsilon \rightarrow$ recommended fortigate change
- with probability $\epsilon \rightarrow$ recommend something new

- what exploration does -
- when youtube shows you a new creator, it collect data:
 - did you click?
 - did you watch full video?
 - did you like / share?
- if engagement is high
 - Q-value of that video increases
 - Algorithm learns.

- How exploration finds viral content -
 - you normally watch tech videos.
 - a new creator uploads an amazing AI documentary
 - without exploration $\rightarrow$ you never see it.
 - with exploration $\rightarrow$ it is tested on small audience.
- If audience engagement spikes -
 - algorithm increases distribution
 - video becomes viral

- why exploration is necessary -
 - Algorithm overages prematurely
 - personalization becomes narrow
 - user experience becomes repetitive
 - platform growth slows.

2) with the help of appropriate example construct the bellman equation and relate to value iteration and policy iteration.

→ Bellman eqⁿ.

- The bellman expresses, "The value of a state = immediately reward + discounted value of next state."

for a given policy π :

$$V^\pi(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s,a) [R(s,a) + \gamma V^\pi(s')]$$

for the optimal value function -

$$V^*(s) = \max_a \sum_{s'} P(s'|s,a) [R(s,a) + \gamma V^*(s')]$$

eg, simple 2-state mdp -

- state $S_1 \rightarrow$ can go to S_2
- state $S_2 \rightarrow$ terminal

31- Bellman construction -

for S_1 - $V(S_1) = R + \gamma V(S_2)$

substitute

$$V(S_1) = 5 + 0.9(0)$$

$$V(S_1) = 5$$

slightly more complex eqⁿ -

from S_1 -

Action $A_1 \rightarrow S_2$ (reward 5)

Action $A_2 \rightarrow S_1$ (reward 1)

Now,

$$V^*(S_1) = \max [5 + 0.9V(S_2), 1 + 0.9V(S_1)]$$

$$= \max [5, 1 + 0.9V]$$

$$\therefore V = 10$$

So,

- Action A1 gives value = 5
- Action A2 gives value = 10

$$V^*(s_1) = 10$$

- Relation to value iteration -

value iteration repeatedly applies the bellman optimality eqⁿ -

$$V_{k+1}(s) = \max_{\alpha} \sum p(s'|s, \alpha) [R(s, \alpha) + \gamma V_k(s')] \quad \eta$$

Process - initialize $V_0(s) = 0$

- Update values using bellman eqⁿ
- repeat until convergence
- extract policy -

$$\pi^*(s) = \operatorname{argmax}_{\alpha} \sum p(s'|s, \alpha) [R + \gamma V(s')]$$

- Relation to policy iteration -

Policy iteration uses two steps -

1- Policy evaluation -

uses bellman expectation eqⁿ -

$$V^{\pi}(s) = \sum p(s'|s, \pi(s)) [R + \gamma V^{\pi}(s')] \quad \eta$$

compute value under current policy

2- Policy improvement -

$$\pi_{\text{new}}(s) = \operatorname{argmax}_{\alpha} \sum p(s'|s, \alpha) [R + \gamma V^{\pi}(s)]$$

Improve policy greedily.

repeat until policy stops changing.